RAJALAKSHMI ENGINEERING COLLEGE

P Saam 251801276-2025-AIDS

P2

Started on	Thursday, 27 August 2026, 9:04 AM
State	Finished
Completed on	Thursday, 27 August 2026, 9:06 AM
Time taken	1 min 23 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Problem Statement:

Given a sorted array and a value x, the floor of x is the largest element in array smaller than or equal to x. Write divide and conquer algorithm to find floor of x.

Input Format

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Value for x

Output Format

First Line Contains Integer – Floor value for x

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>
#include <stdlib.h>

// Divide and Conquer function to find the floor of x
int findFloor(int *arr, int low, int high, int x) {
    int floor_val = -1; // Default value if no floor exists

    while (low <= high) {
        int mid = low + (high - low) / 2;

        // If exact match found, it is the floor
        if (arr[mid] == x) {
            return arr[mid];
        }
        // If element is smaller than x, it's a potential floor
        else if (arr[mid] < x) {
            floor_val = arr[mid];
            low = mid + 1;        // Search in the right half for a closer (larger) floor
        }
        // If element is greater than x, look in the left half
        else {
            high = mid - 1;
        }
    }

    return floor_val;
}

int main() {
    int n;
    // Read the size of the array
    if (scanf("%d", &n) != 1) return 0;

    int *arr = (int *)malloc(n * sizeof(int));

    // Read n array elements (each on a new line as specified)
    for (int i = 0; i < n; i++) {
        scanf("%d", &arr[i]);
    }

    int x;
    // Read the value for x
    scanf("%d", &x);

    // Find the floor value using divide and conquer
    int result = findFloor(arr, 0, n - 1, x);

    // Print the result
    printf("%d\n", result);

    // Free allocated memory
    free(arr);
```

	Input	Expected	Got	
✔	6 1 2 8 10 12 19 5	2	2	✔
✔	5 10 22 85 108 129 100	85	85	✔
✔	7 3 5 7 9 11 13 15 10	9	9	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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